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Date: May 2026

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Effects of heat source arrangements on arc behavior and keyhole stability in Laser-Arc hybrid welding of 12-mm-thick AH36 steel

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Abstract

Laser–Arc Hybrid Welding (LAHW) critically depends on the guiding mode that sets the spatiotemporal sequence of the heat sources. Yet, how the guiding mode governs keyhole stability, the physical core of process robustness, remains poorly understood. Using AH36 high-strength steel plates of 12mm thickness, we systematically compare Arc-Leading (A-L) and Laser-Leading (L-A) modes by combining side-view high-speed imaging (10,000 fps) with synchronous electrical measurements. We show that the two modes establish distinct bidirectional couplings between the electrically driven arc stability and the mechanically driven keyhole stability. In the A-L mode, the leading arc first forms a wide,

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high-temperature molten pool that buffers droplet momentum and suppresses high-frequency keyhole oscillations. After a short free-growth period ($\sim 0.8\text{ms}$), the arc is captured and locked by the laser-formed keyhole, eliminating deflection and markedly increasing arc stiffness. In contrast, the L-A mode lacks such buffering; droplets directly impinge on the rear keyhole wall, periodically inducing partial collapses and a keyhole reconstruction period that is $\approx 1.8\times$ that of the A-L mode. Macroscopically, the A-L mode achieves a stable root at a gap of 0.6mm, whereas the L-A mode requires 0.8mm to suppress humping. These results reveal mode-specific feedback loops—virtuous in A-L and self-exciting in L-A—and provide a mechanistic basis for widening the robust process window in industrial LAHW.

Keywords

Laser-Arc hybrid welding; Guiding mode; Keyhole behavior; Arc characteristics; Process stability

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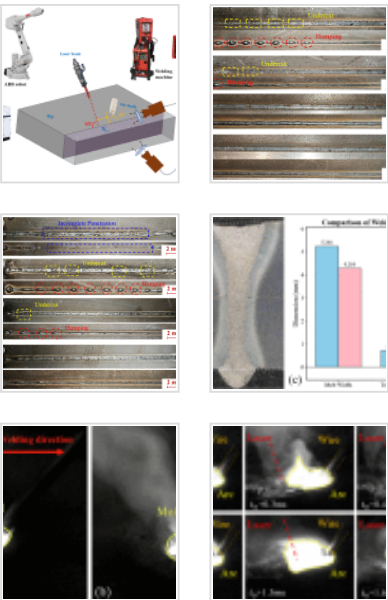
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1. Introduction

In high-end equipment manufacturing sectors, such as aerospace and new energy vehicles, critical load-bearing components are subject to extreme requirements for welding quality and efficiency [1]. Conventional fusion welding has encountered performance bottlenecks, while standalone laser welding or Gas Metal Arc Welding (GMAW) have inherent limitations in terms of process window and penetration capability [2]. As an advanced joining technology, Laser-Arc Hybrid Welding (LAHW) achieves synergistic performance by combining the “deep penetration” of the laser with the “high filling” of the arc [3]. However, the realizing this synergy depends critically on the stability of the welding process. The physical core of process stability lies in the ability to

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form and maintain a stable keyhole that penetrates the molten pool [4]. The dynamic behavior of the keyhole, including its oscillation, collapse, and reconstruction, not only directly determines the energy coupling efficiency [5] but also governs the formation of welding defects such as porosity [6] and humping [7].

Among the numerous process parameters, the laser-arc guiding mode is a fundamental factor in modulating the physical environment of the keyhole [8]. The influence of the guiding mode, which determines the spatiotemporal sequence of the laser beam and the arc—namely Laser-Leading (L-A) or Arc-Leading (A-L)—on process stability has been extensively investigated. However, the findings reveal a complex interplay of competing physical phenomena, and a universally optimal configuration remains elusive [9]. Research has often focused on two primary aspects: the electrical/plasma characteristics [10] and the hydrodynamic/thermal behavior [11], [12]. From an electrical standpoint, the L-A mode is frequently favored for its ability to stabilize the arc. The study by Li et al. [13] confirmed that the laser-preformed keyhole and its metallic vapor plume provide a highly conductive, low-resistance channel for the trailing arc, thereby effectively anchoring it, suppressing deflection, and concentrating its energy. Conversely, from a hydrodynamic and thermal perspective, the A-L mode often demonstrates superiority. The pre-heating from the leading arc establishes a large molten pool, which not only improves the subsequent laser's energy absorption [14] but also promotes a more stable droplet transfer by mitigating the repulsive force from the laser-induced vapor. However, Zhu et al. [15] provided crucial counter-evidence, revealing that the leading arc's dynamic pressure in the A-L mode can deform the molten pool, increasing the keyhole's front wall angle to a less favorable state for laser absorption compared to the near-Brewster angle achievable in the L-A mode, thereby reducing the laser's absorption efficiency.

These insightful studies, however, have predominantly treated the arc behavior and keyhole dynamics as separate, often unidirectional cause-and-effect phenomena. For instance, research has focused on how a stable laser keyhole benefits the arc [16] or how a leading arc affects the molten pool for the laser [17], but the bidirectional feedback loop between these two dynamic elements remains poorly understood. The core question of

how the mechanically-driven keyhole stability (influenced by droplet impact and molten pool buffering, as discussed by Cai et al. [18]) and the electrically-driven arc stability (governed by the conductive path [19], [20]) are dynamically and intrinsically coupled under different guiding modes has not been systematically deconstructed. This knowledge gap in understanding the bidirectional coupling mechanism constitutes a significant theoretical bottleneck, hindering the development of a predictive framework for process stability.

To fill this gap, this study aims to reveal the underlying physics of this bidirectional coupling. It is posited that the significant difference in process stability between the A-L and L-A modes is governed by the distinct nature of the feedback cycle established in each configuration. This research seeks to deconstruct how the mechanically-driven keyhole stability (influenced by droplet impact and molten pool buffering) and the electrically-driven arc stability (governed by the conductive path) interact and influence one another within this coupled system. To this end, we employ high-speed imaging and synchronous electrical signal analysis techniques to systematically deconstruct and compare the real-time dynamic behaviors of both the arc and the keyhole. The objective is to establish the intrinsic relationship between the arc's electrical characteristics and the keyhole's mechanical stability, thereby elucidating the fundamental mechanisms that govern overall process stability and providing a theoretical basis for establishing a highly robust process window for hybrid welding.

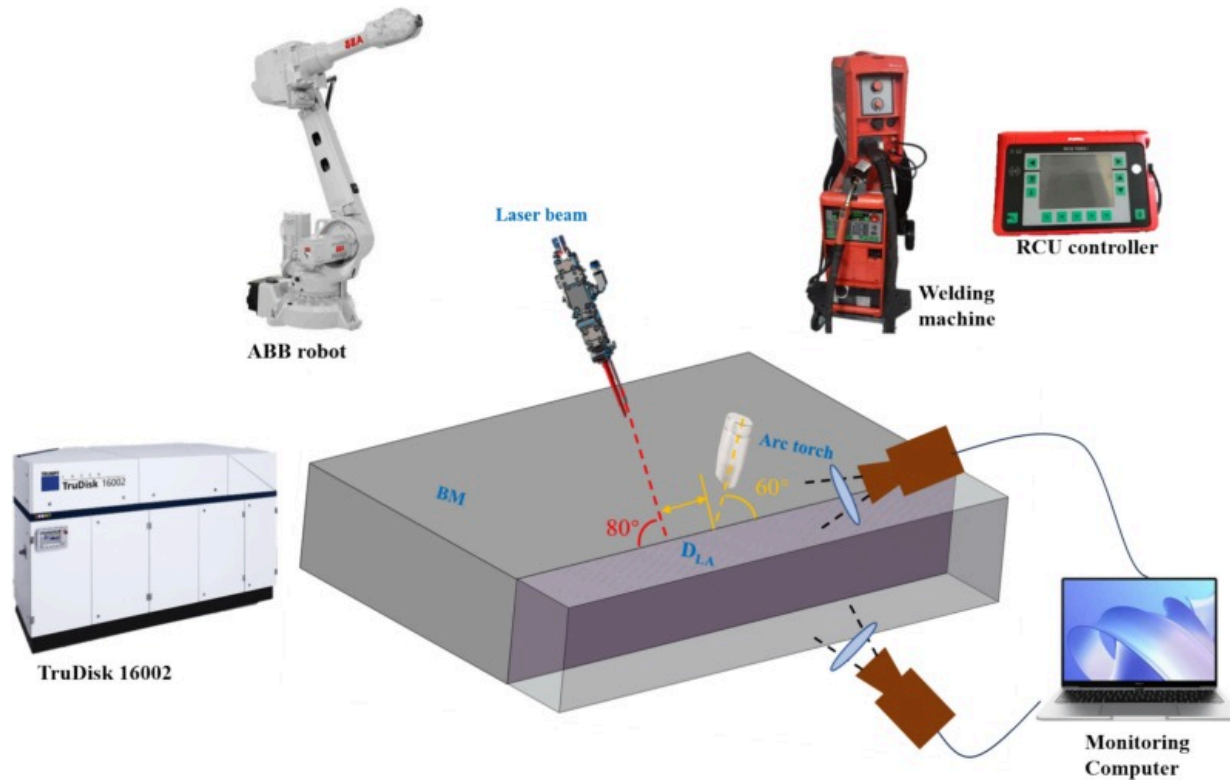
2. Materials and methods

The base material used in this study was AH36 high-strength steel, with dimensions of 300mm×150mm×12 mm. A square butt joint was used, with single-sided welding achieving full penetration and back-bead formation. A G4Si1 high-strength wire with a diameter of 1.2mm was used as the filler material. The chemical compositions of the base material and filler wire are detailed in Table 1. Prior to welding, specimen surfaces were ground to remove oxides and contaminants.

Table 1. Chemical compositions of the base material and filler wire (wt.%).

	C	Si	Mn	P	S	Fe
AH36	0.170	0.10–0.50	0.7–1.6	≤0.035	≤0.035	Bal.
G4Si1	0.088	0.860	1.690	0.012	0.017	Bal.

The experimental setup included a TruDisk 16002 disk laser as the laser source, which has a maximum power of 16kW, a minimum spot diameter of 0.45mm, and a wavelength of 1030nm. The MAG welding system consisted of an ABB six-axis robot (IRB 4600-60/2.05) and a Fronius TPS5000 welding machine. MAG welding was performed in direct-current (DC), non-pulsed mode. To capture process details, a high-speed camera was utilized for side-view observation at a high frame rate of 10,000 fps. A semiconductor laser with a wavelength range of 810 nm was employed as an illumination source, positioned at the side of the weld seam. An 808nm narrow-band filter and a 1/32 neutral density filter were placed in front of the camera lens to filter out excess reflected light during the welding process. During welding, the angle between the welding torch and the laser head was maintained at 37°. The laser-arc distance (D_{LA}), defined as the horizontal distance between the laser beam center and the wire tip projection on the workpiece surface, was fixed at 4mm. A schematic of the experimental setup is shown in Fig. 1.



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Fig. 1. Schematic illustration of the experimental setup.

The fixed welding process parameters were as follows: a laser power of 9kW, a welding speed of 1.8m/min, and a wire feeding speed of 11 m/min. The shielding gas, a mixture of 80% Ar and 20% CO₂, was supplied at a flow rate of 15L/min. Current and voltage were sampled at high frequency and hardware-synchronized with the camera via external triggering; Details regarding sampling rates and filters are provided in the respective signal analysis sections. After welding, cross-sectional specimens were extracted from the joints and prepared for metallographic analysis. The specimens were ground, polished, and then etched with a 4% nital solution (nitric acid in alcohol). The cross-sectional

morphology of the welds was subsequently observed under a super-depth-of-field optical microscope (OM). The specific process parameters are listed in [Table 2](#).

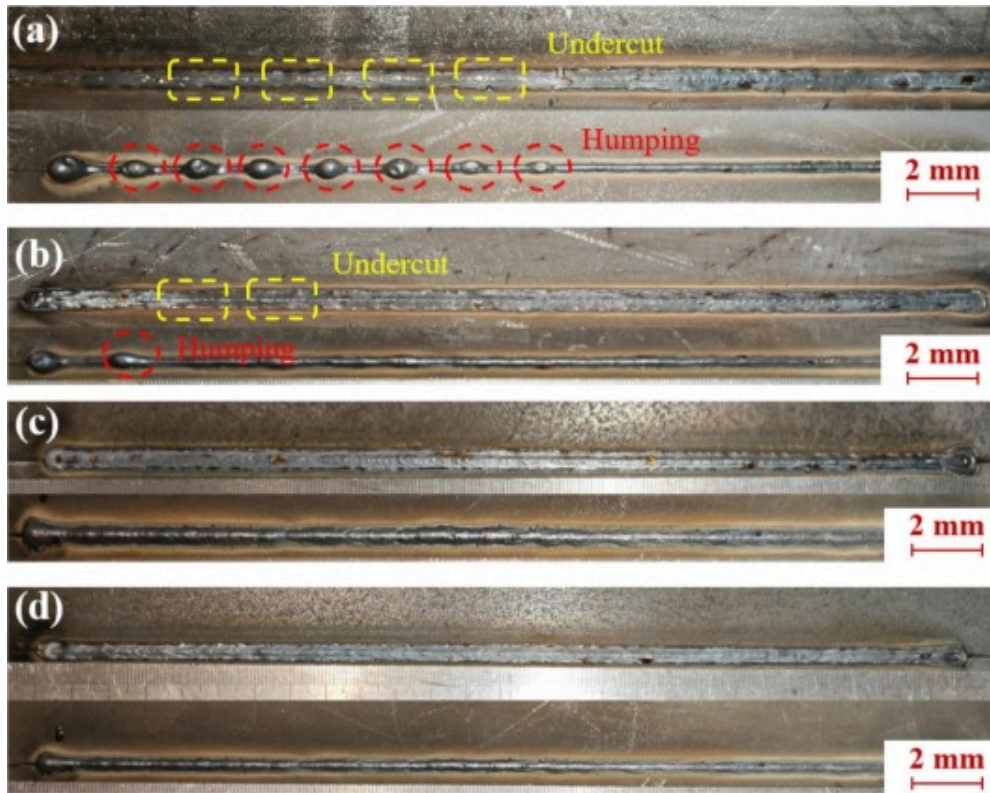
Table 2. Welding parameters used in the experiments.

Parameters	Value
Laser power/kW	9
Defocusing distance/mm	−3
Welding speed/m·min ^{−1}	1.8
Wire feeding speed/m·min ^{−1}	11
Distance of laser and wire/mm	4
Shield gas flow rate/L·min ^{−1}	15
Gap widths/mm	0.2, 0.4, 0.6, 0.8

3. Results and discussion

3.1. Effect of guiding mode on gap sensitivity and weld formation

The macroscopic morphology of the top and root surfaces for welds produced in the A-L mode is presented in [Fig. 2](#). At a narrow gap of 0.2mm ([Fig. 2a](#)), the weld formation was poor: the top bead exhibited periodic collapse and burn-through, while severe humping defects formed on the root in the first half of the weld due to unstable penetration. As the gap was widened to 0.4mm ([Fig. 2b](#)), these defects were alleviated. When the gap reached 0.6mm ([Fig. 2c](#)), defects on both the top and root surfaces were completely eliminated, yielding a continuous and uniform weld bead, which indicated that the penetration behavior had stabilized. This stable state was maintained as the gap was further increased to 0.8mm ([Fig. 2d](#)).



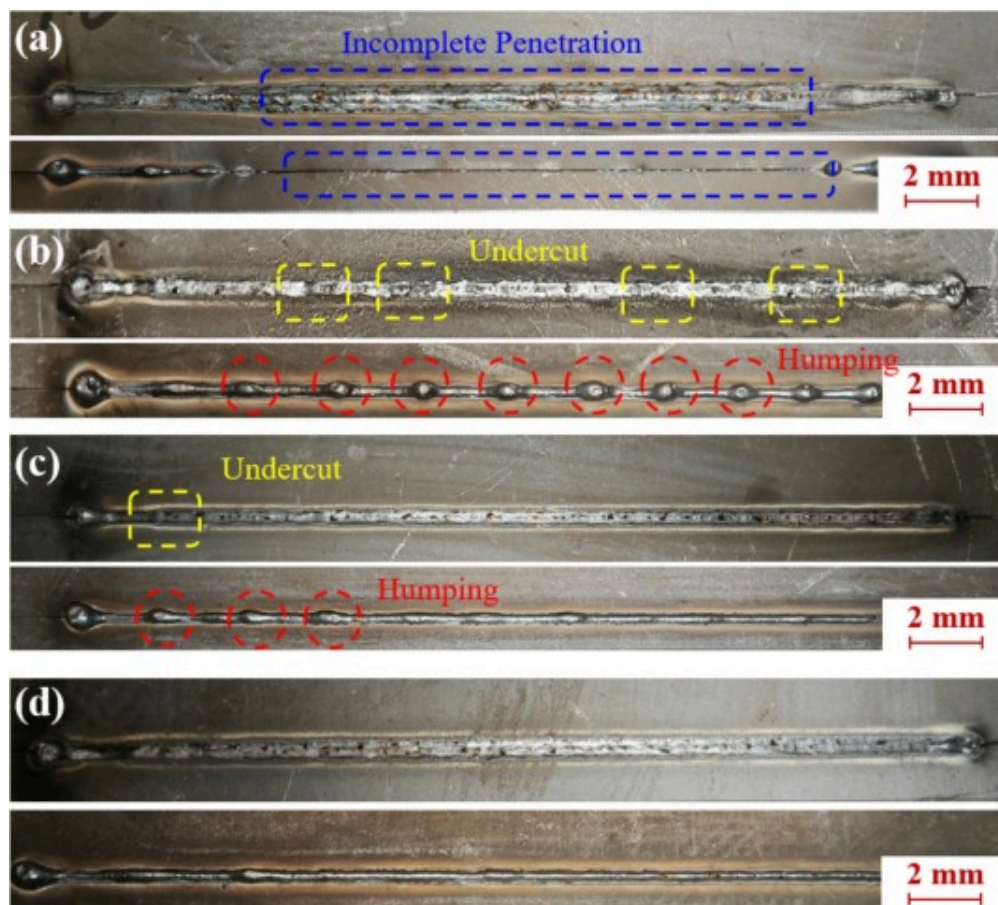
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Fig. 2. Weld formation in the A-L mode under different gap widths: (a) 0.2mm, (b) 0.4mm, (c) 0.6mm, and (d) 0.8mm.

The macroscopic morphology of the top and root surfaces for welds produced in the L-A mode is presented in Fig. 3. At a narrow gap of 0.2mm (Fig. 3a), the weld root exhibited a significant lack of penetration, achieving only intermittent, point-like penetration at the start and end of the seam, which suggests a highly unstable keyhole. Concurrently, the top bead was characterized by excessive reinforcement. When the gap was increased to 0.4mm (Fig. 3b), uniformly distributed humping defects formed along the entire length of the root, and the top bead displayed a distinct wavy profile. This defect pattern indicates a

consistent, periodic instability governed by the process parameters at this gap width. As the gap was further widened to 0.6mm (Fig. 3c), the root humping was partially mitigated, with the defects becoming smaller and more discretely distributed. Nevertheless, the overall weld still exhibited a state of discontinuous penetration, indicating that a stable process window had not yet been reached. A continuous and uniform root pass was finally achieved only when the gap was increased to 0.8mm (Fig. 3d), at which point the top bead also became relatively smooth.



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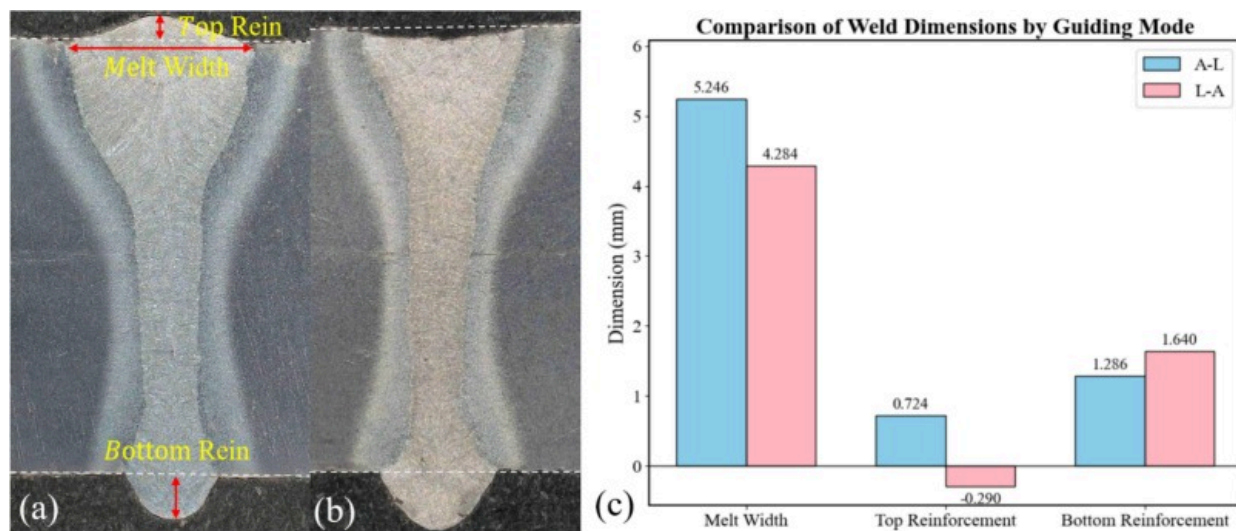
Fig. 3. Weld formation in the L-A mode under different gap widths: (a) 0.2mm, (b) 0.4mm, (c) 0.6mm, and (d) 0.8mm.

A comparison of the experimental results for the two guiding modes reveals a significant difference in their tolerance to the welding gap. The diametrically opposed defect modes observed at the narrow gap of 0.2mm can be attributed to the distinct energy coupling mechanisms of the two modes. In the A-L mode, the leading arc significantly preheats the base material, lowering the viscosity and surface tension of the molten metal. Within the confined 0.2mm gap, the recoil pressure from the trailing laser easily drives this highly fluid melt downward. Restricted by the narrow gap, the fluid cannot spread laterally and instead bulges at the root, causing the observed humping defects [21]. Conversely, in the L-A mode, the narrow gap restricts the entry of the trailing arc. Due to the minimum voltage principle and the blockage effect of the laser-induced plasma plume within the tight gap, the arc preferentially attaches to the top edges of the plates. This results in energy concentration at the top surface while the root receives insufficient heat, leading to incomplete penetration [22]. A stable root formation was achieved in the A-L mode with a 0.6mm gap, whereas the L-A mode exhibited extensive, uniformly distributed humping at the same gap width and only produced a stable weld when the gap was increased to 0.8mm. The reason for this discrepancy lies in the fundamental difference in the sequence of heat source application: in the A-L mode, the leading arc provides wide-area preheating of the base material, creating more favorable conditions for the trailing laser to achieve penetration [23]. Conversely, in the L-A mode, the laser creates a narrow heat-affected zone, and the trailing arc must act on relatively cold base material, thus requiring a larger physical gap to ensure effective energy input [24].

The distribution characteristics of the humping defects also differed distinctly between the two modes. In the A-L mode, humping defects were concentrated at the initial section of the weld (Figs. 2a, b). This can be attributed to the significant heat accumulation effect from the leading arc; as welding progressed, the temperature of the base material continuously increased, improving the fluid dynamic conditions of the molten pool and thereby suppressing subsequent hump formation [14]. In contrast, the humping defects in

the L-A mode were uniformly distributed along the entire length of the weld ([Fig. 3b](#)). This is because the thermal effect of the leading laser is localized, resulting in negligible heat accumulation along the weld path. Consequently, the physical conditions remained largely consistent throughout the process, and once the parameter combination induced humping, the defect persisted along the entire seam. In this study, the weld length on the base material was 300mm. However, a potential implication for industrial applications involving multi-meter welds is that the significant heat accumulation in the A-L mode might induce thermal expansion and gap variations. Nevertheless, this does not imply that the stability of the A-L mode is confined to short welds. In long-distance welding, the thermal field eventually transitions from a transient accumulation phase to a thermal equilibrium. The challenge of gap variation is primarily a fixture-related engineering issue rather than an intrinsic process instability. Therefore, while the A-L mode aids in rapidly achieving stability during the startup phase via preheating, its robust process window remains advantageous in long welds, provided that proper clamping or adaptive parameter control is employed to manage global thermal expansion.

To further compare the differences in joint formation, representative cross-sections of well-formed joints from both modes were selected for analysis ([Fig. 4](#)). The statistical results revealed that the A-L mode produced a greater fusion width and sufficient weld reinforcement. In contrast, the L-A mode resulted in a narrower fusion width and an underfill morphology on the weld face.



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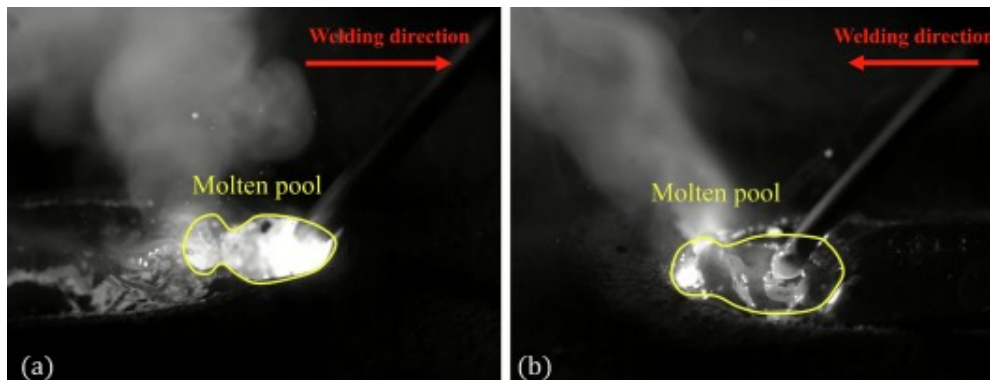
Fig. 4. Joint morphology and dimensional statistics under different guiding modes: (a) joint cross-section for the A-L mode; (b) joint cross-section for the L-A mode; (c) comparison of dimensional statistics.

This underfill can be attributed to the heat source sequence specific to the L-A mode. The leading laser pre-melts the base material, and the trailing arc, governed by the minimum voltage principle [25], preferentially initiates deep within this pre-existing molten pool. This “buried arc” mode of droplet transfer results in the majority of the filler metal being deposited inside the pool, failing to effectively build up the weld reinforcement. Consequently, this leads to an underfill on the weld face.

3.2. Influence of guiding mode on molten pool and arc morphology

To investigate the fundamental influence of the guiding mode on the morphology of the molten pool and arc, the resulting pool shapes were captured via high-speed photography, as presented in Fig. 5. In the A-L mode (Fig. 5a), the leading arc first

establishes a wide initial molten pool. This pool is subsequently elongated and stirred by the trailing laser, which ultimately forms a single, bullet-shaped molten pool with a smooth posterior. In contrast, the process in the L-A mode (Fig. 5b) is markedly different. The keyhole-associated molten pool created by the leading laser is relatively short and narrow. Consequently, the trailing arc cannot fully envelop it; the front of the arc plasma merely overlaps with the rear of the laser-induced pool. These two largely separate molten pools interact via the backward jet from the keyhole and the transfer of droplets, leading to the formation of a unique, gourd-shaped composite molten pool.



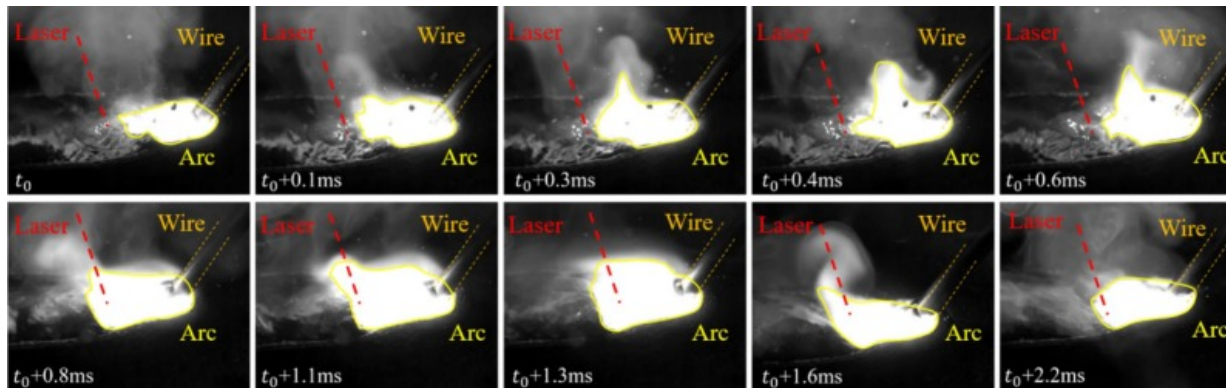
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Fig. 5. Comparison of molten pool morphology under different guiding modes: (a) A-L mode; (b) L-A mode.

To investigate the evolution of arc morphology under different guiding modes, its periodic behavior was captured using high-speed photography. The cyclical evolution of the arc morphology in the A-L mode is presented in Fig. 6. At time t_0 , the arc exhibited a diffuse morphology. During the period from $t_0+0.3\text{ms}$ to $t_0+0.6\text{ms}$, the arc root began to deflect, as the relatively small arc was influenced by the laser plasma. At $t_0+0.8\text{ms}$, the arc expanded and extended over the keyhole formed by the trailing laser in the molten pool, which generated an electromagnetic attraction on the arc. This attraction increased the arc's stiffness and anchored its position at the keyhole opening, causing the deflection to

cease. The arc subsequently remained stable for the period of $t_0 + 0.8\text{ms}$ to $t_0 + 2.2\text{ms}$. In this locked state, the droplet transfers onto the liquid front wall of the keyhole. This liquid interface acts as a buffer, effectively absorbing the impact momentum and preventing direct disruption to the keyhole stability.



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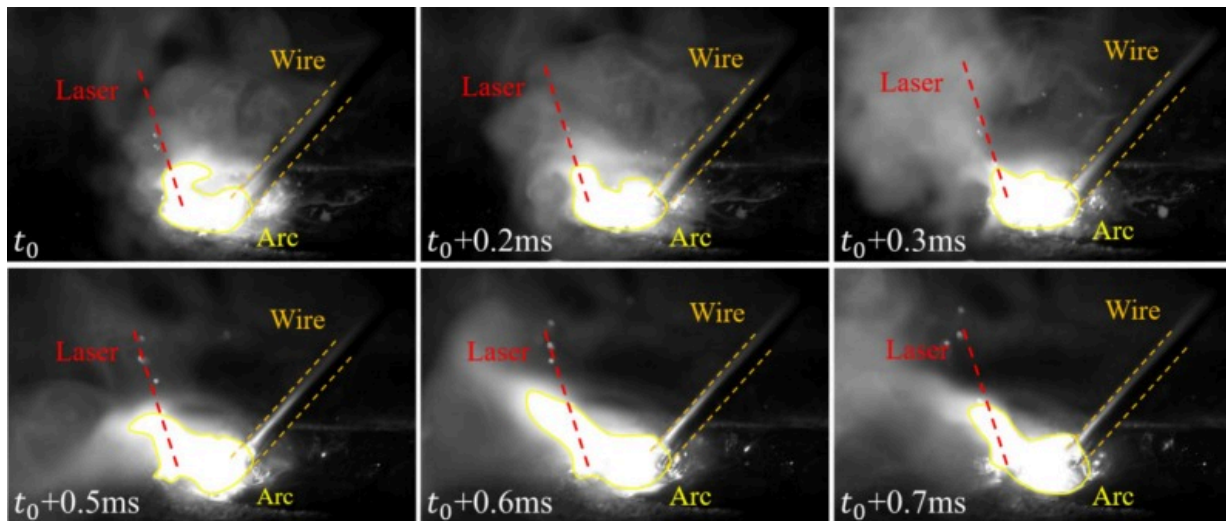
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Fig. 6. Arc morphology in the A-L mode.

This process reveals that in the A-L mode, the arc acts as a leading heat source and undergoes a brief, unconstrained “free-growth period” of approximately 0.8ms after ignition. During this phase, the arc's volume and the momentum of its plasma jet can fully develop. However, being unconstrained, it is prone to free deflection. When this mature, high-stiffness arc traverses into the interaction zone of the trailing laser, it is attracted and captured by the laser-induced keyhole. Because the arc's intrinsic stiffness is already substantial at this stage, once it is locked in place, it can effectively resist external perturbations, leading to prolonged stable burning. Therefore, a key characteristic of the A-L mode is this transient “free-growth period”; once the arc is fully developed and subsequently locked by the keyhole, it achieves a state of extremely high stability. It is worth noting that the universality of this mechanism is subject to specific boundary conditions involving welding speed and power input. The formation of the stabilizing buffer pool is time-dependent; at significantly higher welding speeds [26], the

leading arc may lack sufficient time to melt the base material adequately, which would diminish the buffering capacity and potentially destabilize the process. Similarly, the duration of the “free-growth period” is intrinsically linked to the arc current. Under lower power conditions, the electromagnetic stiffness of the arc decreases, making it more susceptible to external disturbances and potentially prolonging the unstable free-growth phase before effective capture occurs [27].

The periodic evolution of the arc morphology in the L-A mode is presented in Fig. 7. At time t_0 , the arc appeared diffuse. By $t_0+0.3\text{ms}$, its morphology had contracted, and the burning process began to stabilize. As the arc volume expanded into the interaction zone of the leading laser, the laser-generated plasma plume exerted a clamping effect on the arc. This constraining force restricted the arc, allowing it to maintain stable combustion during the period from $t_0+0.5\text{ms}$ to $t_0+0.7\text{ms}$.



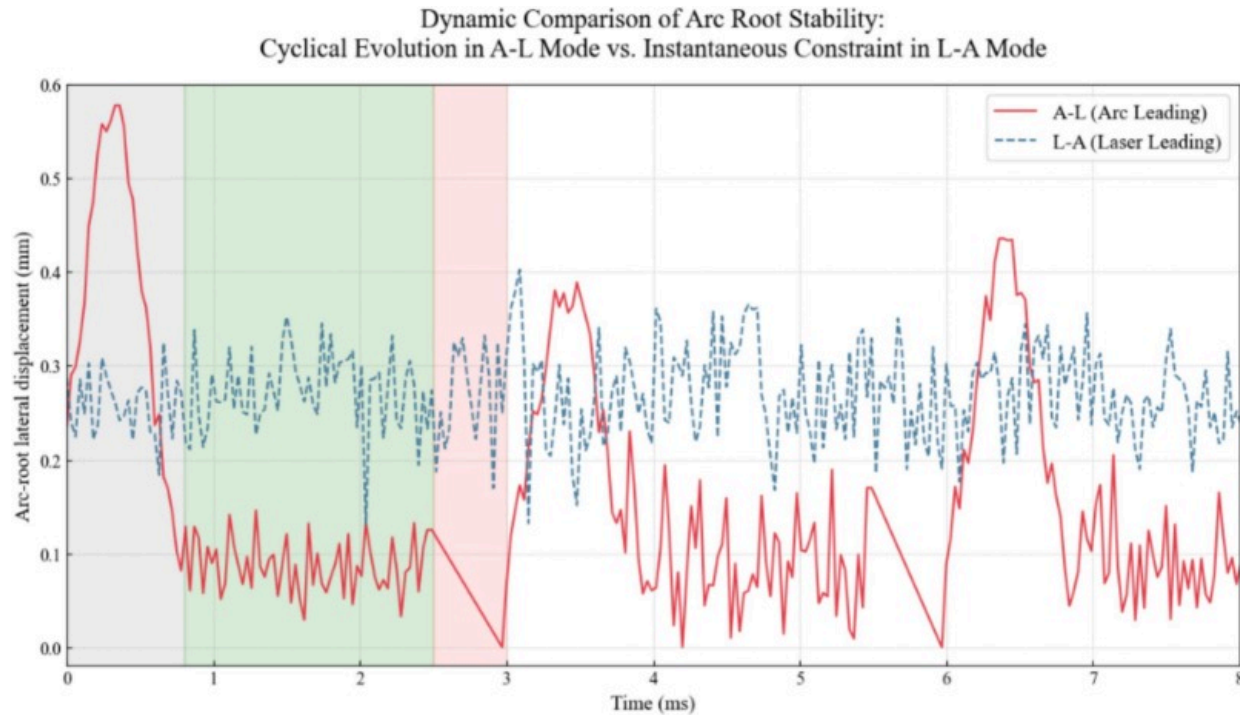
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Fig. 7. Arc morphology in the L-A mode.

In this mode, the arc ignites in a region already pre-heated by the laser and filled with conductive metal plasma. Consequently, it has virtually no “free-growth period.” Instead, while its stiffness and volume are still in a nascent stage, it is immediately subjected to a strong clamping and guiding effect from the leading laser's plasma plume. This prevents the arc from developing freely or deflecting; it is rapidly constrained to a relatively stable morphology and position. Therefore, a key characteristic of the L-A mode is that the arc is stable almost immediately upon ignition, but its morphology and energy distribution are, to a much greater extent, dictated by the leading laser.

The curves were obtained by frame-by-frame tracking of the arc-root position from side-view high-speed imaging, with pixel-to-mm calibration and short-window smoothing to derive the time history of lateral arc-root displacement, as Fig. 8. The background colors highlight a single representative A-L cycle: gray denotes the 0–0.8ms free-growth stage (arc unconfined, large lateral excursions), green the 0.8–2.2ms capture-lock stage, and pink the 2.2–2.6ms quench/collapse stage. It is important to clarify that the near-zero displacement observed during this pink stage (2.2–2.6ms) does not imply that the arc is stabilized at the center; rather, it indicates the short-circuit transfer phase where the arc momentarily extinguishes. During this interval, the arc column vanishes, and the displacement is mathematically defined as zero in the tracking algorithm until re-ignition occurs. Accordingly, the A-L trace (red solid) shows a pronounced transient peak at the start of each cycle, followed by a low-amplitude stable band and a near-zero segment in the pink window. In contrast, the L-A trace (blue dashed) remains on a low-frequency baseline around 0.09–0.11 mm without sharp spikes, indicating instantaneous laser clamping and stronger geometric confinement. The figure thus visualizes the distinct temporal architectures of arc–keyhole coupling: a periodic “free-growth→lock→quench/re-ignite” rhythm in A-L versus a persistently stable baseline in L-A.



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Fig. 8. Arc displacement over time under different guiding modes.

The electrical behavior of the arc in LAHW, under the steady-state or quasi-steady-state assumption where transient effects are neglected, is governed by the current continuity equation for the electric potential field distribution. Its general form is given by Eq. (1) [28]:

$$\nabla[\sigma(x)\nabla\phi]=0 \quad (1)$$

where ∇ is the nabla operator, representing a differential operation on spatial coordinates (in this context, divergence). σ is the electrical conductivity of the plasma, which varies with the spatial position x . ϕ is the electric potential. The gradient of this potential field determines the electric field strength $\mathbf{E} = -\nabla\phi$, which in turn defines the spatial

distribution of the current density j via Ohm's Law: [29]:

$$\mathbf{j} = \sigma \mathbf{E} = -\sigma \nabla \phi \quad (2)$$

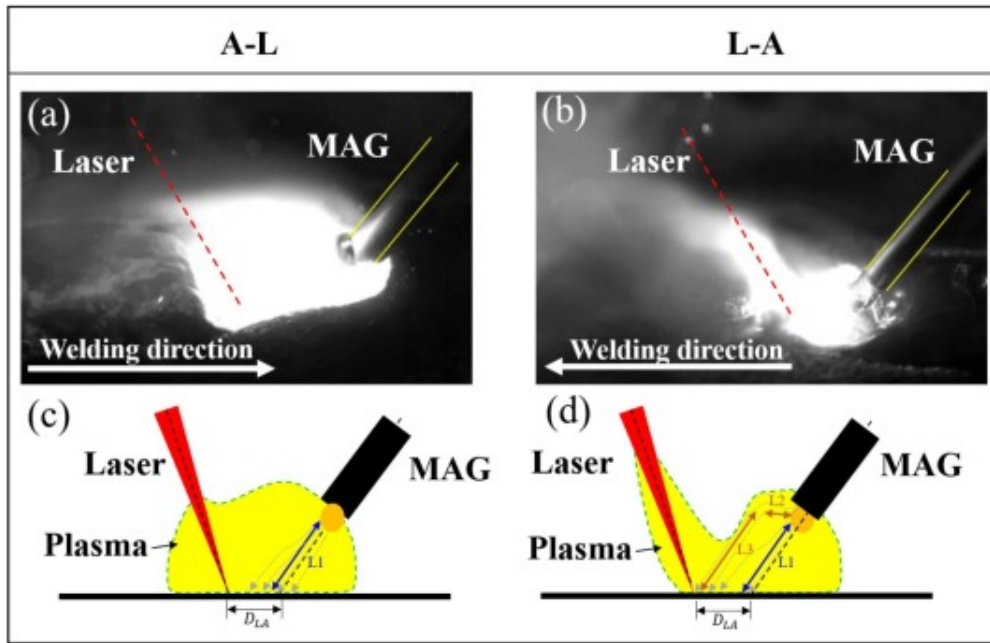
Here, $\mathbf{j}$ is the current density vector, and $\mathbf{E}$ is the electric field vector. The physical meaning of this set of equations is that the current lines will spontaneously rearrange in space to seek the path of minimum total power dissipation. The ohmic dissipation per unit volume (i.e., Joule heating) is given by $\mathbf{q} = \mathbf{jE} = \mathbf{j}^2 / \sigma$. The system tends to minimize the overall dissipation, $\int \mathbf{q} d\mathbf{v}$, which is macroscopically equivalent to the current selecting the path of “minimum equivalent resistance. The key physical parameter that dictates this path selection is the electrical conductivity, σ . The conductivity of the plasma is extremely sensitive to temperature and the degree of ionization, a dependency confirmed in the literature. Based on plasma physics, its relationship can be approximately described by Eq. (3) [30]:

$$\sigma \simeq \frac{n_e e^2}{m_e \nu_m} \quad (3)$$

Here, n_e is the electron density, which increases with the degree of ionization; e is the elementary charge; m_e is the electron mass; and ν_m is the momentum collision frequency between electrons and heavy particles (ions and neutral atoms). Because an increase in electron temperature T_e significantly enhances the degree of ionization (thereby increasing n_e), the electrical conductivity σ increases exponentially with temperature. As a result, the arc channel will spontaneously close along the highest conductivity region composed of high-temperature, highly ionized metal vapor, even if this path is geometrically longer.

In the A-L mode, the arc first forms a wide liquid anode region at the leading edge, producing significant arc dynamic pressure and electromagnetic pinch, as Fig. 9 (a). This pushes the frontal liquid column back toward the keyhole opening, increasing the inclination angle of the keyhole's front wall, which in turn weakens the initial laser absorption and the plume's conductivity. Under these conditions, σ is highest at the workpiece surface while the plume ahead is insufficient, satisfying the condition

$R_{L1} < R_{L2+L3}$, Consequently, the arc stabilizes and spreads along the L1 path (the shortest surface channel) [31].



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Fig. 9. Influence of the guiding mode on arc characteristics and the conductive path.

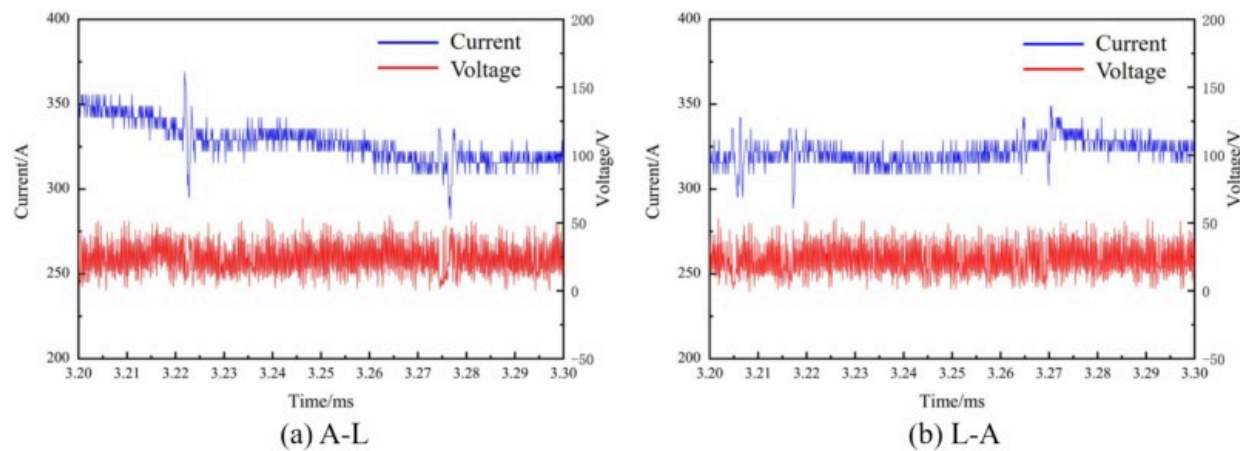
Conversely, in the L-A mode, the laser first establishes the keyhole and produces a strong plume at the front wall. Its initial absorption can be described by the Fresnel model [32]:

$$A(\beta) = 1 - R_s(\beta) - R_p(\beta) \quad (4)$$

When the front wall incident angle β approaches the “quasi-Brewster condition” for ferrous metals, the absorptivity $A(\beta)$ is maximized, resulting in the strongest plume [33]. The front wall angle in the L-A mode is approximately 81.6° – 84.7° , whereas in the A-L mode, it is about 87.3° [15]. Consequently, in the L-A mode, σ_{plume} rises rapidly, causing the condition $R_{L2+L3} < R_{L1}$ to be met. The arc first ignites along the surface path L1 and

is subsequently guided by the highly conductive plume. The conductive channel then transitions to the L2 (upward along the plume)+L3 (toward the keyhole/cathode region) path, exhibiting a trajectory change of “L1→L2+L3”, as Fig. 9 (b). This enhancement of the conductive channel by laser-induced metal vapor has been identified as a key mechanism for improving arc stability.

The raw transient electrical signals, presented in Fig. 10, completely record the full dynamic information of the process under both guiding modes. However, due to the high complexity and stochastic nature of these fluctuations, it is difficult to directly and quantitatively discern the differences in process stability between the two modes by observation of the raw data alone.



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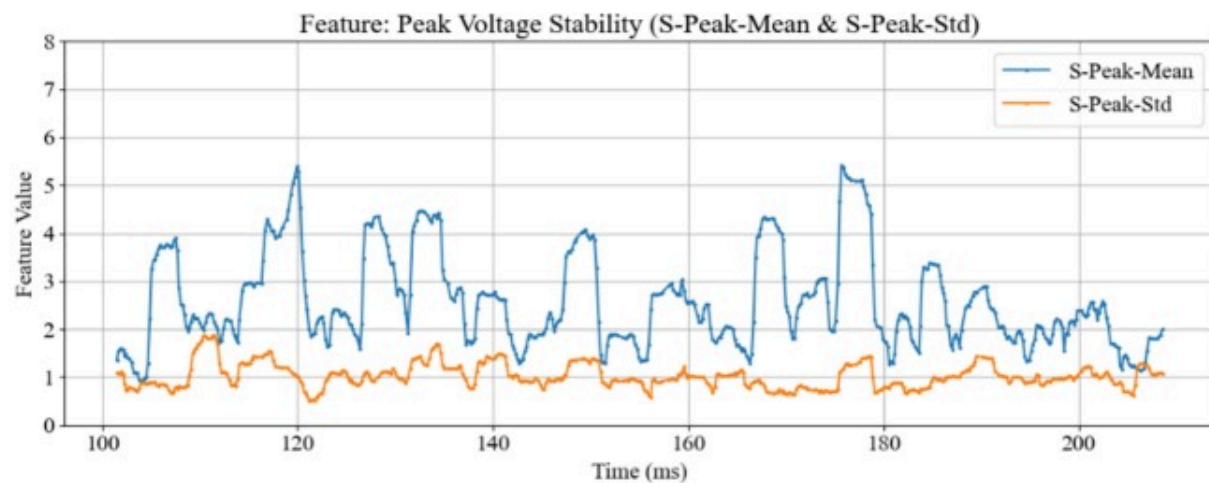
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Fig. 10. Current and voltage waveforms of the Laser-MAG process under different guiding modes: (a) A-L mode; (b) L-A mode.

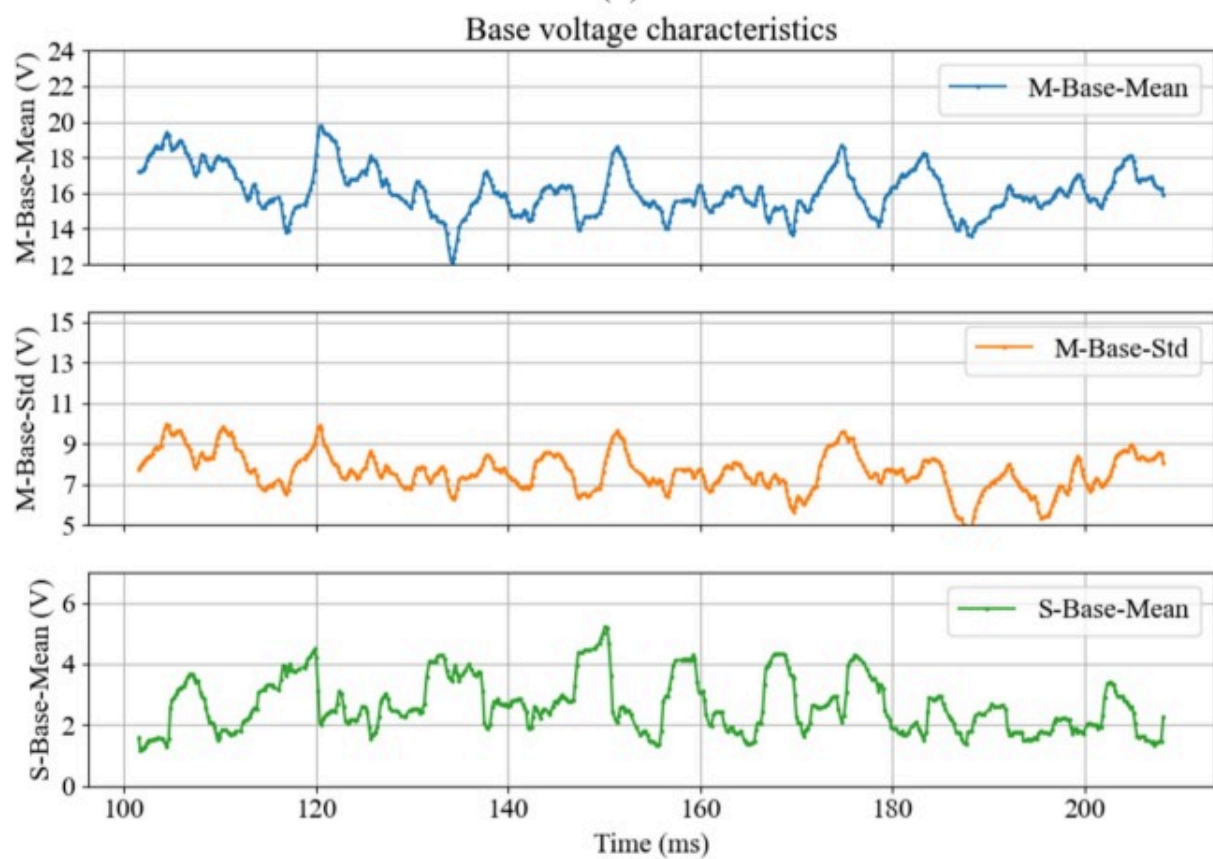
To enable in-depth analysis and quantitative characterization of process stability from the high-frequency voltage signals, a statistical analysis method based on a sliding window was introduced [34]. The statistical features presented in Fig. 11 and Fig. 12 were derived using a sliding window analysis method to quantitatively characterize process stability

from the high-frequency voltage signal. First, after identifying and smoothing the local mean (p'_{mean}) and standard deviation (p'_{std}) of each voltage peak, a main sliding window with a width of $N=30$ peaks is applied to calculate the collective stability indicators. The peak voltage features are intended to quantify the stability and consistency of droplet transfer. S-Peak-Mean is the mean of the per-pulse peak mean values within the sliding window, representing the average voltage magnitude required for droplet transfer. Its calculation formula is shown in Eq. (5):

$$S\text{-Peak-Mean} = \sqrt{\frac{1}{N} \sum_{i \in W} (m_{P,i} - \bar{m}_P)^2} \quad (5)$$



(a)

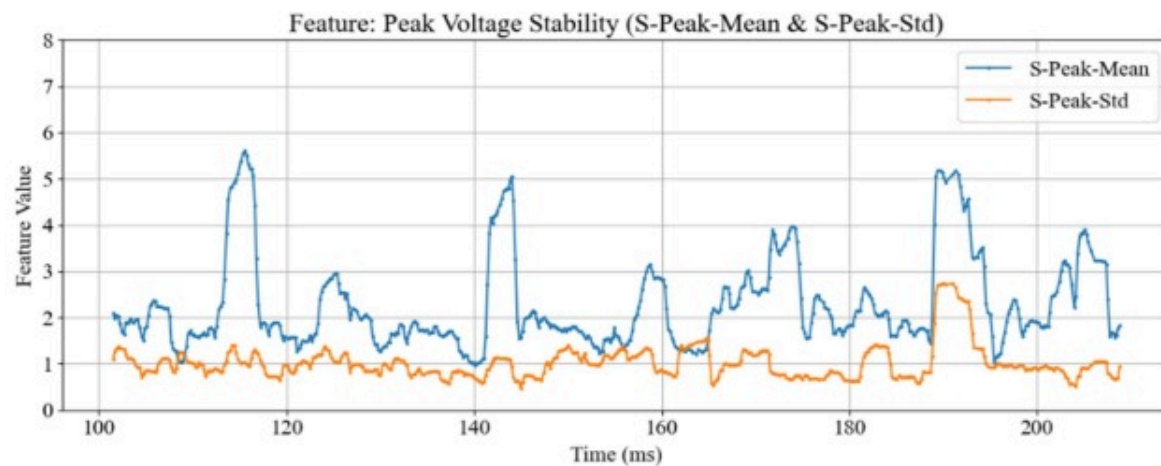


(b)

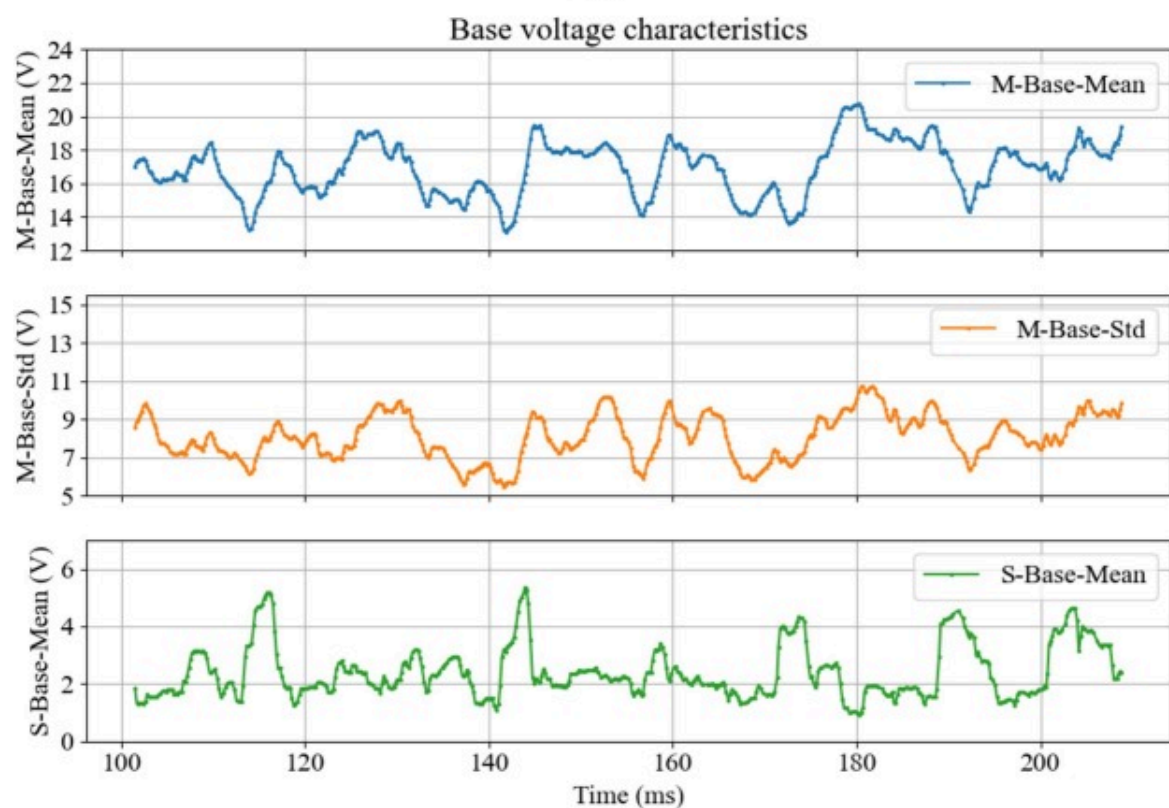
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Fig. 11. Voltage stability characteristics in the A-L mode: (a) stability of the peak voltage;
(b) characteristics of the base voltage.



(a)



(b)

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Fig. 12. Voltage stability characteristics in the L-A mode: (a) stability of the peak voltage; (b) characteristics of the base voltage.

A higher value typically signifies that the arc morphology is more unstable and elongated. S-Peak-Std (the collective mean of peak standard deviations) is the mean of the per-pulse peak standard deviations within the window and is used to quantify the consistency of the transfer process. Its calculation formula is shown in Eq. (6):

$$S\text{-Peak-Std} = \sqrt{\frac{1}{N} \sum_{i \in W} (s_{P,i} - \bar{s}_P)^2} \quad (6)$$

A lower value indicates that the droplet transfer is more uniform and the process is more stable. The base voltage features are used to characterize the basic burning state of the arc. M-Base-Mean is the mean of the per-pulse base mean values (Base-Mean) within the window, and this indicator is directly related to the average arc length. Its calculation formula is shown in Eq. (7):

$$M\text{-Base-Mean} = \frac{1}{N} \sum_{i \in W} (\text{mean}(B_i)) \quad (7)$$

M-Base-Std is the mean of the per-pulse base standard deviations (Base-Std) within the window, representing the stability of the base voltage within a single pulse period. A higher value indicates more severe arc disturbances. Its calculation formula is shown in Eq. (8):

$$M\text{-Base-Std} = \frac{1}{N} \sum_{i \in W} (\text{std}(B_i)) \quad (8)$$

S-Base-Mean is the standard deviation of the per-pulse base mean values (Base-Mean) within the window, representing the stability of the arc length across multiple pulse periods. Its calculation formula is shown in Eq. (9):

$$S\text{-Base-Mean} = \sqrt{\frac{1}{N} \sum_{i \in W} (\text{mean}(B_i) - M\text{-Base-Mean})^2} \quad (9)$$

The stability characteristics of the peak voltage in the A-L mode are presented in [Fig. 11a](#), the S-Peak-Std curve is not constant but exhibits periodic “humps”. Physically, these humps indicate that when the arc deflects and lengthens, the droplet transfer mode deteriorates. The high standard deviation implies that the droplets are no longer detaching uniformly; instead, the transfer becomes irregular, likely shifting towards a repelled transfer mode due to the distorted electromagnetic forces in the deflected arc. A key feature is that the stability indicators, S-Peak-Mean and S-Peak-Std, are highly synchronized with the fluctuations of the main arc's base voltage (M-Base-Mean), as shown in [Fig. 11b](#). During the 'free deflection' phases, when the base voltage is elevated (e.g., at 110ms and 135ms), both the mean and standard deviation of the peak voltage synchronously reach their maxima. This indicates that the voltage required for droplet transfer is higher and less consistent during these periods. Conversely, when the arc is 'captured' by the keyhole and the base voltage decreases, these indicators also drop to their nadir, and the droplet transfer process regains stability. This cyclical variation clearly reflects the physical process of the arc: the rise and fall of the base voltage correspond to the 'free deflection' and 'capture-lock' phases, respectively.

In contrast, [Fig. 12\(a\)](#) reveals a distinct transfer mechanism. The S-Peak-Std maintains a low, flat baseline, signifying a highly stable, one-pulse-one-droplet transfer mode governed by the laser's constraint. However, distinct sharp spikes are observed (e.g., at $t=118\text{ms}$). These spikes serve as electrical fingerprints for discrete short-circuit events or anomalous droplet detachments. Unlike the continuous instability in the A-L mode, the L-A mode is characterized by overall stability punctuated by these rare, instantaneous short-circuit defects. Its S-Peak-Mean and S-Peak-Std indicators remain at a stable, low baseline for the vast majority of the time, punctuated by instantaneous, sharp, pulse-like spikes at specific moments (e.g., at 118ms and 142ms). This “stable baseline with transient spikes” signature stands in stark contrast to the prolonged, “hill-shaped” phases of instability observed in the A-L mode. This behavior can be attributed to the relatively stable base voltage of the L-A mode ([Fig. 12b](#)). Under the continuous constraint of the leading laser, the arc length is effectively stabilized, preventing any significant deflection.

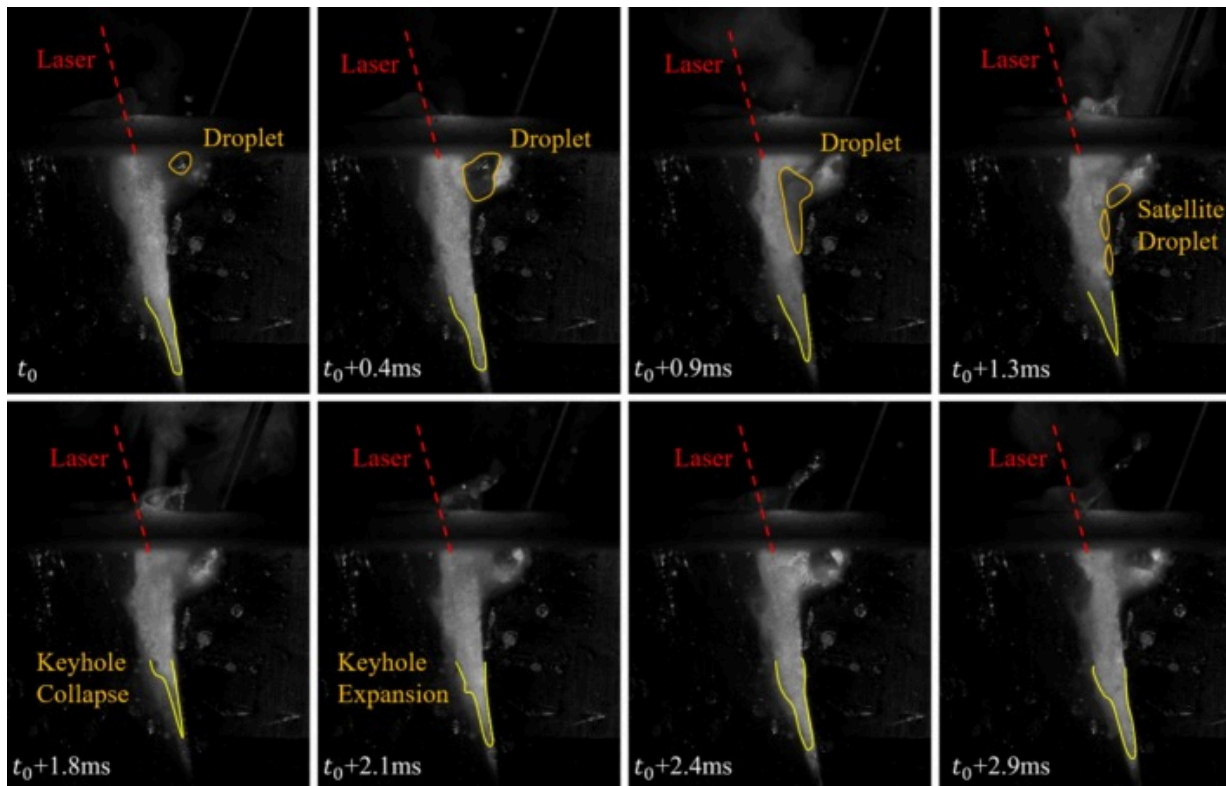
Therefore, the process in this mode is generally stable, with instability primarily originating from sporadic, discrete disturbance events, such as non-axial droplet transfer. Once such a disturbance ceases, the system immediately recovers its stability under the strong constraint of the laser. As a result, the duration of these instability signals is extremely short.

This fundamental difference in arc stabilization dynamics is not an isolated electrical phenomenon. As will be detailed in the following section, the ability of the arc to achieve and maintain a stable state is intrinsically dependent on the mechanical stability of the keyhole, which serves as the physical and electrical anchor for the arc plasma.

3.3. Dominating effect of guiding mode on keyhole behavior and stability

Due to the different sequence of heat source application on the base material, the evolution of the laser-induced keyhole also varies between the guiding modes. To investigate the principles of this evolution, the keyhole dynamics during the welding process were captured from a side-view perspective using a sandwich method. Fig. 13 shows the evolution of the keyhole over a droplet transfer cycle in the A-L mode. At time t_0 , the formation and transfer of a droplet are driven by electromagnetic force and surface tension. At $t_0+0.9\text{ms}$, the droplet transfers into the molten pool, and its impact disrupts the keyhole's original mechanical equilibrium, causing the lower part of the keyhole to expand. From $t_0+1.3\text{ms}$ to $t_0+2.1\text{ms}$, the droplet's energy dissipates within the keyhole, inducing hydrodynamic oscillations of the keyhole wall. The tendency for the keyhole to close is suppressed when the laser-induced metal vapor recoil pressure, which keeps the keyhole open, exceeds the sum of the molten pool's surface tension and hydrostatic pressure. This dominant opening force effectively counteracts the closing tendency caused by the droplet impact and drives the redistribution of molten metal on the keyhole wall. This process facilitates the restoration of the keyhole's profile from an irregular, transient state to a more stable, inverted-cone structure. By $t_0+2.4\text{ms}$, the keyhole's depth, opening width, and wall morphology have essentially recovered to a

quasi-steady state, similar to that before the droplet transfer, thus completing one cycle of keyhole evolution influenced by a single droplet.



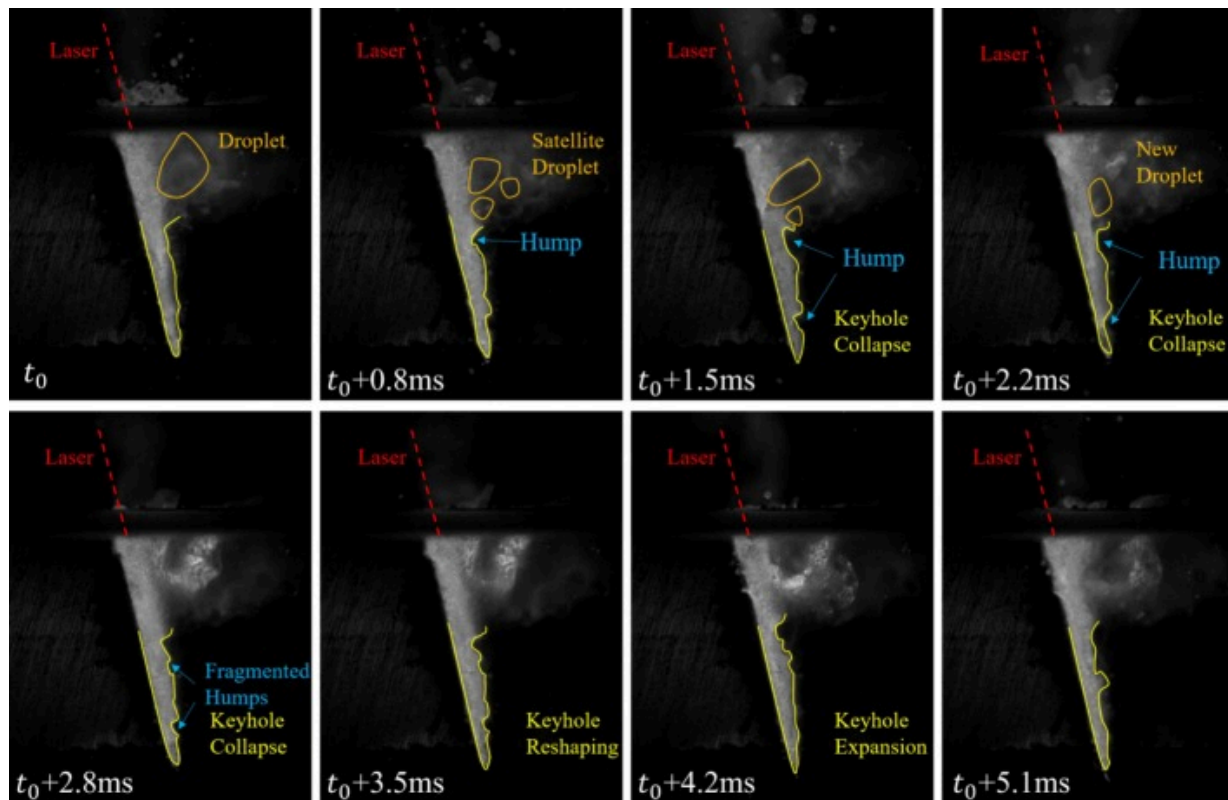
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Fig. 13. Dynamic evolution of the keyhole in the A-L mode.

The evolution of the keyhole during a droplet transfer cycle in the L-A mode is depicted in Fig. 14. The process begins with a droplet impinging on the unstable rear keyhole wall. Subsequently, from $t_0+0.8\text{ms}$ to $t_0+1.5\text{ms}$, the additional dynamic impact pressure generated by the droplet transfer disrupts the force balance on the wall, leading to a large-scale inward collapse of the wall toward the keyhole's center and causing severe channel constriction. At $t_0+2.2\text{ms}$, the instability, dominated by the inward collapse of the

rear wall, results in a periodic, partial collapse of the keyhole, which blocks the transmission of laser energy to the bottom of the joint. During the period from $t_0+3.5\text{ms}$ to $t_0+4.2\text{ms}$, the continuous heating of the collapsed region by the laser beam generates a powerful recoil pressure through metal re-vaporization. This recoil pressure, which is the key force for keeping the keyhole open, ultimately expels the liquid metal and reopens the channel.



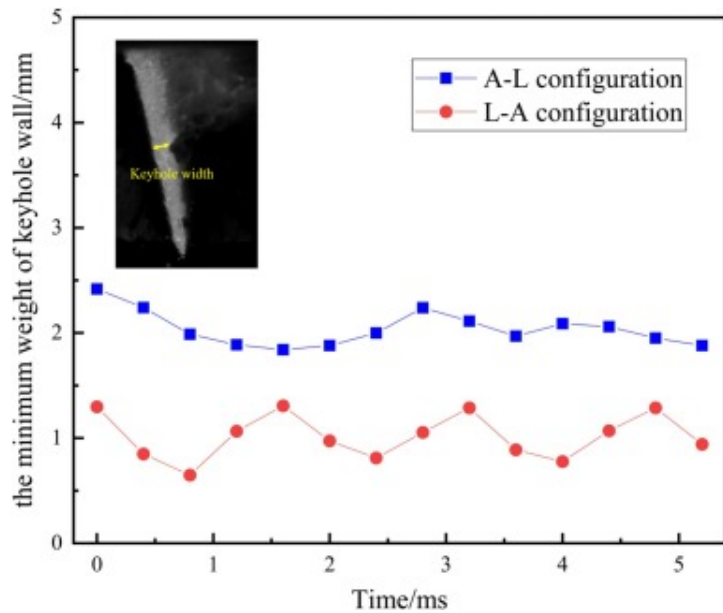
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Fig. 14. Dynamic evolution of the keyhole in the L-A mode.

High-speed videography and subsequent statistical analysis were used to obtain the fluctuation patterns of the minimum keyhole width over a typical cycle for each guiding

mode, as presented in Fig. 15. In the A-L mode, the keyhole exhibited not only a larger average width but also significantly smaller fluctuation amplitudes, demonstrating superior dynamic stability.



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Fig. 15. Keyhole width under different guiding modes.

This enhanced stability can be primarily attributed to two synergistic mechanisms. First, the preheating and arc pressure from the leading arc reduce the surface tension of the material, creating a more favorable condition for the trailing laser to maintain a wide keyhole [35]. Second, the large, high-heat-capacity molten pool established by the arc effectively buffers and suppresses the inherent high-frequency pressure oscillations of the keyhole [36]. A wider and more stable keyhole is a prerequisite for achieving a high-quality weld; it not only facilitates the escape of gases to reduce porosity but also prevents spatter caused by keyhole collapse, thereby ensuring continuous penetration [37].

The choice of guiding mode fundamentally determines the mechanical environment and dynamic stability of the rear keyhole wall. The state of forces on the rear keyhole wall [38] is determined by a balance between the outward metal vapor recoil pressure (F_v), which results from the intense evaporation of metal at the keyhole surface, and the inward closing forces. These closing forces include the metal vapor friction (F_f), a shear force caused by the high-velocity plasma flow along the wall; the molten pool pressure (P_m), which is the hydrostatic pressure exerted by the surrounding liquid metal; the surface tension (F_s), a force that seeks to minimize the surface area and close the keyhole cavity; and gravity (G), which acts on the mass of the molten metal. Among these, the recoil pressure F_v is the key force responsible for maintaining the keyhole's shape, and its strong positive correlation with the keyhole wall surface temperature T_s is defined by Eq. (10) [18]:

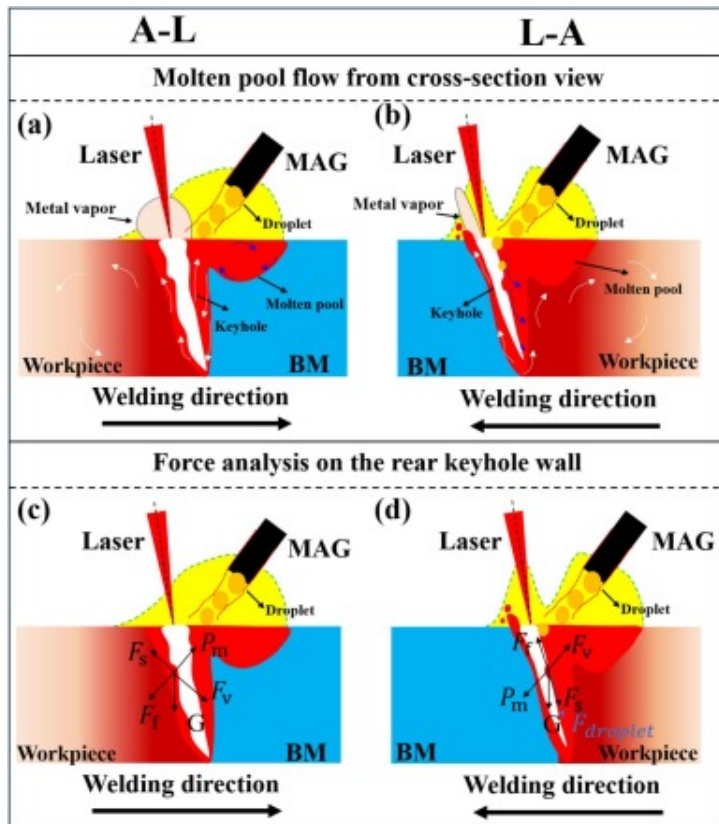
$$F_v = \frac{N_0 B_0}{\sqrt{T_s}} \exp\left(-\frac{m_a H_v}{N_a k_B T_s}\right) \quad (10)$$

Here, N_0 is a numerical coefficient, B_0 is the evaporation constant, T_s is the keyhole wall surface temperature, m_a is the atomic mass, H_v is the latent heat of evaporation, N_a is the Avogadro constant, and k_B is the Boltzmann constant. The recoil pressure F_v is thus highly dependent on the keyhole wall surface temperature. An ideal, quasi-static mechanical equilibrium, free from external disturbances, can be represented by Eq. (11) [39], which describes the balance of forces on the rear keyhole wall:

$$F_f + F_v + F_s + P_m + G = 0 \quad (11)$$

The A-L mode constructs an inherently stable mechanical system. As illustrated in the molten pool flow schematic in Fig. 16(a), the leading arc pre-forms a large molten pool, which allows the impact momentum and energy of the droplet to be dissipated within the fluid via viscous forces and pressure wave propagation, rather than acting directly on the keyhole's mechanical equilibrium boundary [40]. Consequently, the mechanical environment of the rear keyhole wall is effectively isolated from the primary disturbance source of droplet impact. The force analysis, depicted in Fig. 16(c), shows that the system

is free from the $F_{droplet}$ impact term. The keyhole's mechanical behavior thus returns to the quasi-static equilibrium described by Eq. (11). Under this equilibrium, a stable and powerful recoil pressure (F_v) can effectively overcome the conventional closing forces, thereby maintaining a keyhole structure with a smooth profile and stable depth.



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Fig. 16. Influence of the guiding mode on the keyhole opening and the forces acting on the rear keyhole wall.

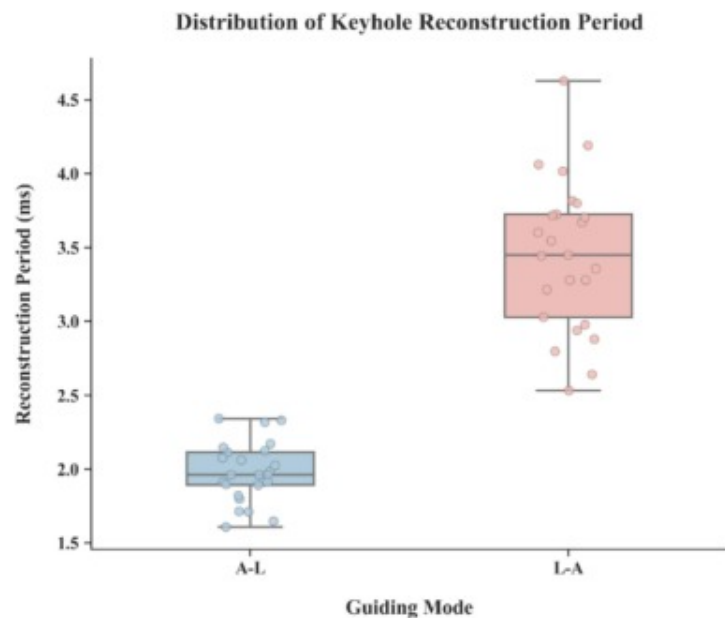
In contrast, the molten pool flow in the L-A mode is shown in Fig. 16(b). Due to the lack of a buffering molten pool, the trailing arc's droplet travels along a trajectory aimed directly at the rear keyhole wall. Upon contact, its momentum is converted into a massive impact

force ($F_{droplet}$), which acts directly on the rear wall [41], as shown in the force analysis in Fig. 16(d). The presence of $F_{droplet}$ causes the system's force state to instantaneously and violently deviate from the equilibrium condition defined by Eq. (11). This impact not only causes severe keyhole oscillations but also leads to the formation of a hydrodynamic “hump” on the rear wall [42]. The protruding part of this hump absorbs more intense laser irradiation, causing the local T_s to rise sharply. According to Eq. (10), this temperature spike generates an anomalously strong local recoil pressure F_v . However, this F_v now has a negative effect: it forcefully pushes the liquid metal at the top of the hump downwards, exacerbating the closing tendency of the middle and lower parts of the keyhole. The complex state of forces under this condition is described by Eq. (12). This positive feedback mechanism—initiated by the $F_{droplet}$ impact and amplified by the anomalous F_v —is the root cause of keyhole instability in the L-A mode:

$$F_f + F_{droplet} + F_s + P_m > F_v \quad (12)$$

The essential difference in the keyhole reconstruction period between the two modes originates from their distinct “molten pool-keyhole” coupling mechanisms. In the A-L mode, the leading arc establishes a large primary molten pool, whose liquid-liquid interface can effectively absorb and dissipate the impact momentum of the droplets, thereby suppressing severe disturbances. This buffering effect of the molten pool on droplet-induced waves has been previously reported [43]. Consequently, keyhole reconstruction involves only rapid pressure rebalancing, resulting in a short period.

The dynamic behavior of the keyhole following droplet disturbances was quantitatively analyzed through frame-by-frame statistical analysis of the high-speed videography. As depicted in Fig. 17, the keyhole reconstruction period in the A-L mode exhibited both a shorter average duration and a more concentrated data distribution, indicating a rapid and stable recovery after perturbation. In contrast, the reconstruction period in the L-A mode was not only significantly longer—measured in this study to be approximately 1.8 times that of the A-L mode—but also considerably more dispersed, which visually reflects the highly unstable nature of the keyhole.



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Fig. 17. Statistical distribution of the keyhole reconstruction period under different guiding modes.

This essential difference in the reconstruction period originates from their distinct “molten pool-keyhole” coupling mechanisms. In the L-A mode, the impact momentum of the droplet superimposes with the fluid momentum of the trailing arc’s own molten pool at the unstable interface between the two pools. When this combined pressure exceeds a stability threshold, it triggers a structural collapse of the rear keyhole wall. This phenomenon of keyhole collapse driven by dynamic impact forces is a known instability mechanism in deep penetration welding. The reconstruction process for such a collapse must rely on a high-energy laser re-vaporization to be completed, leading to the observed prolonged reconstruction period.

Synthesizing the observations of both arc and keyhole dynamics reveals two distinct, self-reinforcing feedback loops that govern the stability of each guiding mode. In the A-L mode, a virtuous cycle is established: the leading arc's pre-formed molten pool provides mechanical buffering, which ensures a stable keyhole. This mechanically stable keyhole, in turn, offers a perfect, low-resistance anchor that enables the arc to achieve the highly stable “capture-lock” state. The stabilized arc then further maintains a consistent keyhole. In contrast, the L-A mode is trapped in a vicious cycle: the lack of a buffer leads to direct droplet impact, causing mechanical instability and periodic keyhole collapse. This fluctuating, unstable keyhole fails to provide a consistent anchor for the trailing arc, leading to the observed electrical instabilities (transient voltage spikes). This unstable arc, in turn, is incapable of helping to stabilize the keyhole. It is this fundamental difference in the bidirectional, mechano-electrical coupling that dictates the overall process stability.

4. Conclusion

Based on a systematic investigation of the mechanisms by which the laser-arc guiding mode dominates the process stability of hybrid welding, and through a comprehensive analysis of weld formation, arc morphology, electrical signal characteristics, and keyhole dynamic behavior, the following main conclusions were drawn:

1. The process window of the A-L mode is significantly superior to that of the L-A mode. The A-L mode demonstrated a wider gap tolerance, achieving a high-quality, stable weld bead at a gap of 0.6mm. In contrast, the L-A mode was highly sensitive to the gap, requiring an increase to 0.8mm to substantially eliminate defects such as inconsistent penetration, indicating a much narrower process window.
2. The two guiding modes exhibit fundamentally different arc stabilization mechanisms and instability characteristics. The arc in the A-L mode undergoes a “free growth → capture-lock” dynamic process. The wide, long-period oscillations in its voltage signal correspond to the “free deflection”

phase before the arc is captured by the keyhole. Once captured, the arc enters a highly stiff and stable state. Conversely, the arc in the L-A mode is “clamped” by the laser upon ignition and cannot develop freely. Its instability does not manifest as long-period deflection but rather as transient, sharp, pulse-like disturbances on an otherwise stable voltage baseline.

3. The A-L mode is more conducive to building a stable and impact-resistant keyhole. In this mode, the large molten pool established by the leading arc acts as a critical “buffer layer,” resulting in a keyhole with a larger average width, smaller fluctuations, and the ability to recover rapidly from droplet impacts. This buffering of dynamic impacts by the molten pool is a recognized mechanism for enhancing stability. In the L-A mode, however, droplets directly impinge on the relatively fragile keyhole wall, readily triggering its periodic collapse, a phenomenon attributed to the dynamic impact force breaking the force balance of the molten pool. Its reconstruction must rely on a high-energy laser re-vaporization process, resulting in a prolonged recovery period approximately 1.8 times that of the A-L mode.

CRediT authorship contribution statement

Lingxiao Song: Writing – original draft, Methodology. **Peilei Zhang:** Supervision, Funding acquisition. **Zufa Li:** Formal analysis. **Junbo Feng:** Conceptualization. **Jixuan Li:** Writing – review & editing. **Yuze Huang:** Investigation. **Chao Fang:** Investigation.

Funding

This research was supported by the National Natural Science Foundation of China grant no. ([52075317](#), [52305435](#)), Class III Peak Discipline of Shanghai—Materials Science and

Engineering (High-Energy Beam Intelligent Processing and Green Manufacturing).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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